Hulle_Madeline_Individual_Assignment_05

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Set Up

Packages

```
# general use and cleaning

library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

#survey data (reading different sheets in as seperate data frames)

aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information

sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quaity

water_quality <- read_xlsx(
   here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
   sheet = "Water Quality",
   # values in df that are actually indicating missing values,
   # making it seem as a chr when it should be num
   # different ways missing values show up in data set
   na = c("over", "", "n/a", "N/A", "173+")
)
```

Class Code

2. Cleaning

`sites` object

```r
# creating new object from sites
ncos_sites <- sites |>
  # filtering to only include NCOS quarterly sites
  filter(sector == "NCOS" &
           sampling_frequency == "Quarterly")
```

clean taxon_list

```r
# creating new object from taxon_list
taxon_list_clean <- taxon_list |>
  # converted taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

clean water_quality

```r
# creatign new object from water_quality
water_quality_clean <- water_quality |>
  # cleaning column name
  clean_names() |>
  # a filtering join, only keep the rows in wq clean in which the
  # site matches up with the site in ncos clean
  semi_join(ncos_sites, by = "site") |>
  # creating new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # grouping by date/site
  group_by(date_site) |>
  # calculating the medians
  summarize(med_pH = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungrouping df
  ungroup() |>
  # filtering out outliers
  filter(date_site != "2022-08-12_NVBR")
```

clean aq_ins

```r
# create a new object
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join only include quarterly ncos surveys
  semi_join(ncos_sites, by = "site") |>
```

```r
# renaming date_on_vial column to date
rename(date = date_on_vial) |>
# selecting columns of interest
# keeping information about each survey and the 9 most abundant taxa
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filtered to only include filteres beaker samples
# (better sampling technique)
filter(sample_type %in% c("FB 250", "FB250")) |>
# converting to long format, making df longer
pivot_longer(
  # name the columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for old column names
  # column names are all in new column called taxon
  names_to = "taxon",
  # new column for the values in each of the old columns
  # counts now under a column called abundance
  values_to = "abundance"
) |>
# group by date site and taxon
group_by(date, site, taxon) |>
# calculated average count per liter (total abundance/7.7)
summarise(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungrouping data frame
ungroup() |>
# new column to join data frames, didnt remove them
unite("date_site", date, site, remove = FALSE) |>
# joined with water quality clean
left_join(water_quality_clean, by = "date_site") |>
# joined with taxonomic information
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# fill in full site names
mutate(site_full = case_when(
```

```
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "East Channel",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # set factor levels for full site name by median salinity
  mutate(site_full = fct_reorder(site_full,
                                  med_sal,
                                  .fun = "median",
                                  .na_rm = TRUE),
         # reversing the order
         site_full = fct_rev(site_full))
```

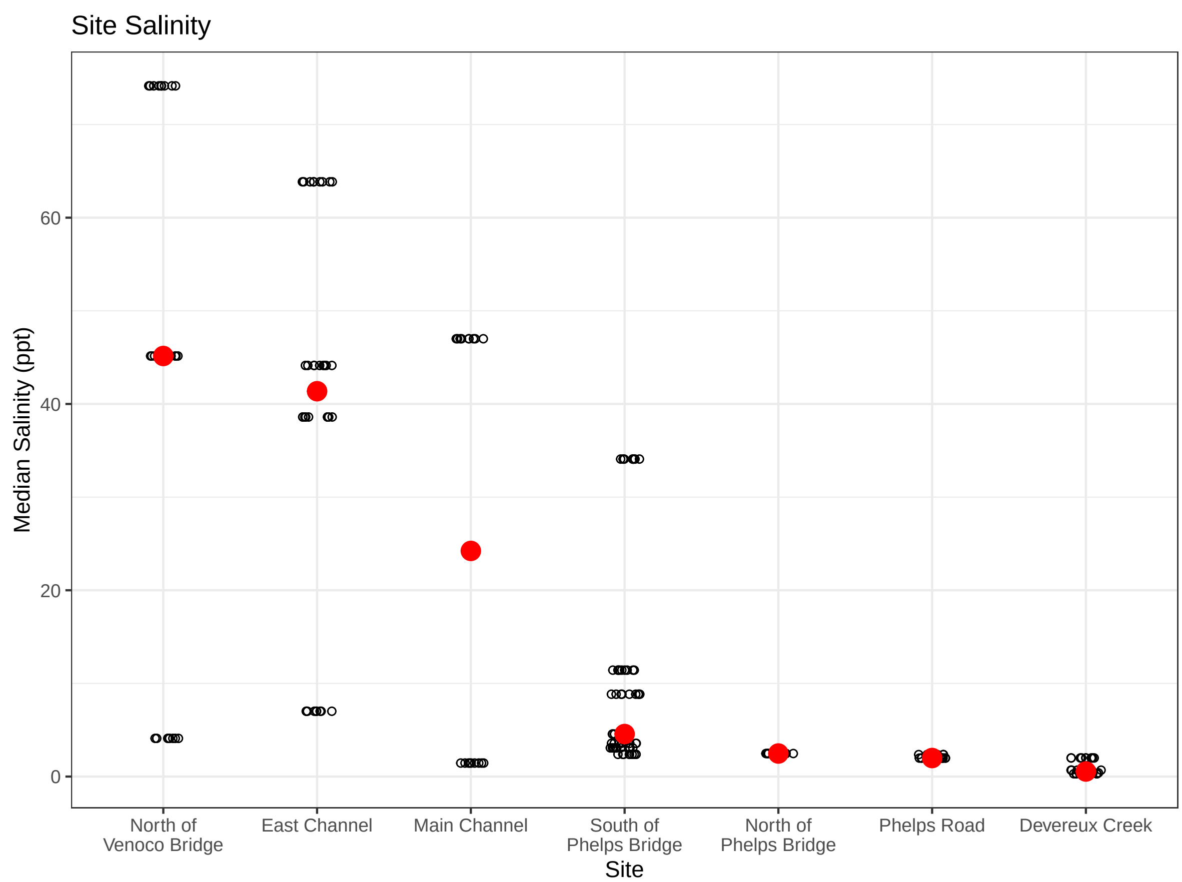
Visualizing

Salinity

```
# creating new objcect to store plot
salinity_plot <- ggplot(data = aq_ins_clean,
                        mapping = aes(x = site_full,
                                      y = med_sal)) +
  # points representing median salinity at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # points representing mean salinity
  stat_summary(geom = "point",
               fun = median,
               color = "red",
               size = 4) +
  # adding line breaks to x axis labels (wrapping)
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # relabeling axes and adding a title
  labs(x = "Site",
       y = "Median Salinity (ppt)",
       title = "Site Salinity") +
```

```r
  # adding themeing
  theme_bw()

# visualizing the object
salinity_plot
```



Copepod Abundance

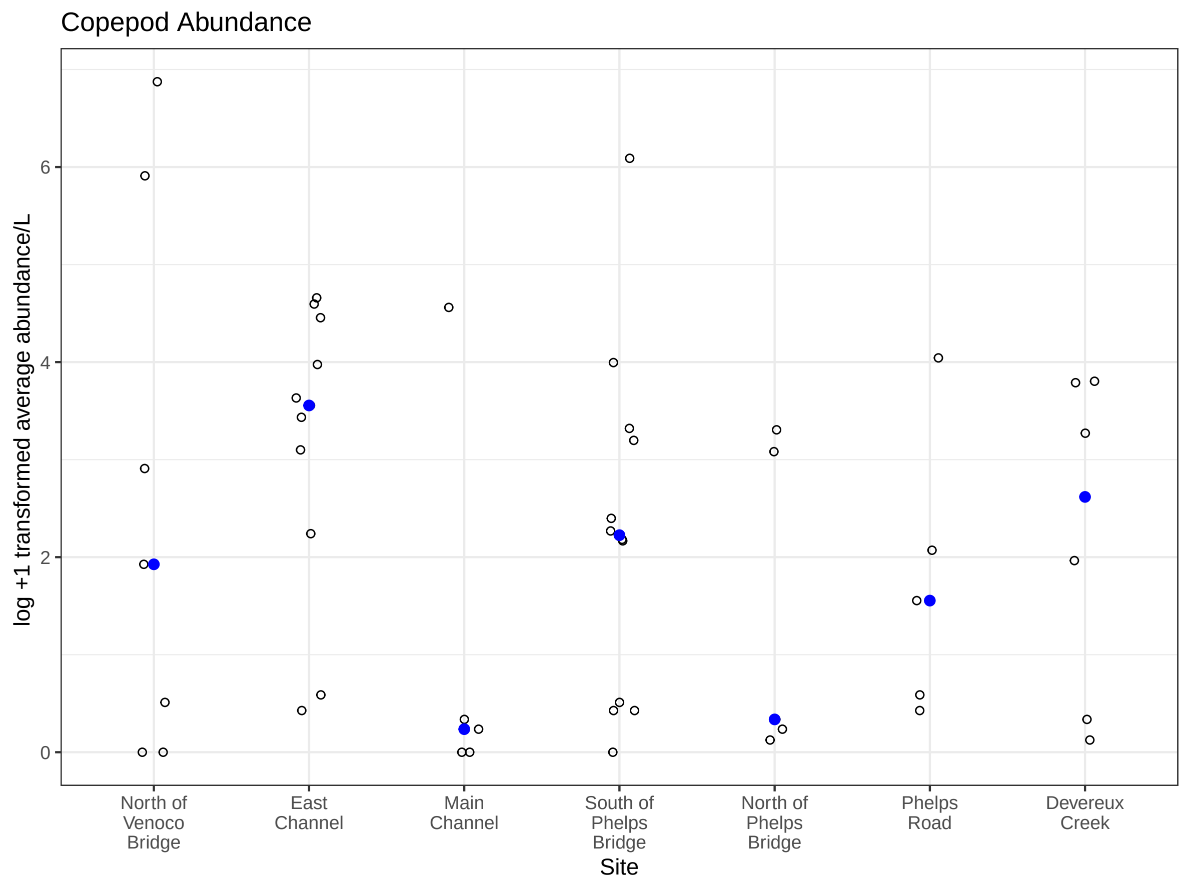
Cleaning

```r
copepods <- aq_ins_clean |>
  filter(taxon == "copepod") |>
  # adding 1 bc some observations are 0
  # helps overcome issues with large differences in order of magnitude
  mutate(log_ave_lit = log(ave_lit + 1))
```

Copepod Viz

```
# creating object to store plot
copepod_plot <- ggplot(data = copepods,
                       mapping = aes(x = site_full,
                                     # using log transformed count per liter
                                     y = log_ave_lit)) +
  # jitter to represent each sample
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # points to represent median copepod abundance
  stat_summary(geom = "point",
               fun = median,
               color = "blue",
               size = 2) +
  # wrappign x-axis labels
  scale_x_discrete(labels = label_wrap(width = 10)) +
  # relabeling axes
  labs(x = "Site",
       y = "log +1 transformed average abundance/L ",
       title = "Copepod Abundance") +
  # using different theme
  theme_bw()

# displaying the object
copepod_plot
```



Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: `site_full`
- y-axis: `log_ave_lit` (updated w/ `log_ave_lit` after viz)
- geometry to represent average abundance/L: geom_jitter()
- geometry to represent medians: stat_summary(geom = "point", fun = median)

b. Wrangle your data

```r
# creating a new object `ostracods` from `aq_ins_clean`
ostracods <- aq_ins_clean |>
  # filtering to only include ostracods
  filter(taxon == "ostracod") |>
  # log transforming `ave_lit` to overcome large difference in
  # order of magnitude between observations
  mutate(log_ave_lit = log(ave_lit + 1))

# taking a sample of 10 observations
slice_sample(ostracods, n = 10)
```

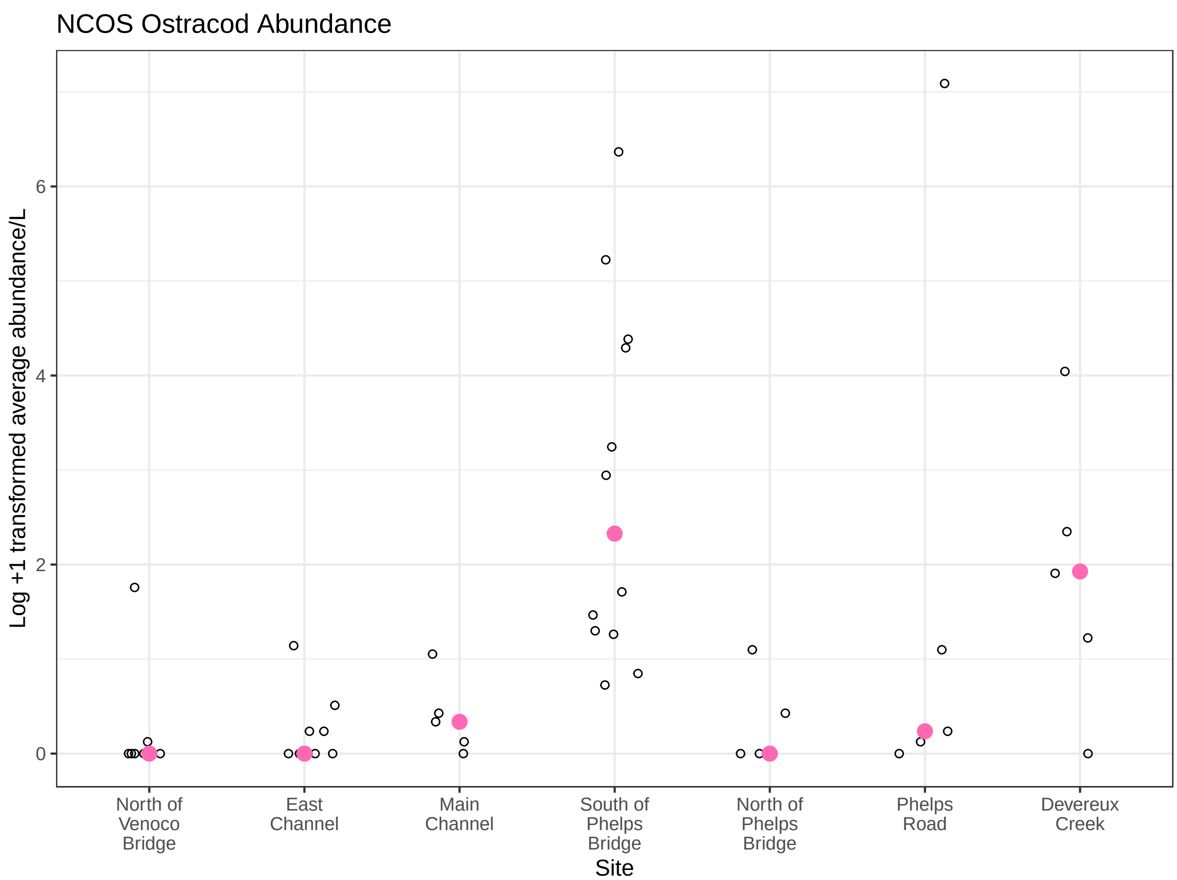
```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_pH med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2023-05-05_NPB  2023-05-05 00:00:00 NPB   ostr~  24.7    NA      NA      12.1
 2 2022-05-11_NDC  2022-05-11 00:00:00 NDC   ostr~   0       9.11    0.398   9.4
 3 2022-11-19_NDC  2022-11-19 00:00:00 NDC   ostr~   6      NA      NA      NA
 4 2022-08-12_NEC  2022-08-12 00:00:00 NEC   ostr~   0       7.37   NA      NA
 5 2023-02-28_NPB1 2023-02-28 00:00:00 NPB1  ostr~   0      NA      NA      NA
 6 2022-05-05_NEC  2022-05-05 00:00:00 NEC   ostr~   0.267   9.45   63.8     7.3
 7 2021-11-23_NPB2 2021-11-23 00:00:00 NPB2  ostr~   0.267   7.59    1.99    0.65
 8 2020-01-21_NDC  2020-01-21 00:00:00 NDC   ostr~   2.4     6.53    0.3     7.6
 9 2023-02-12_NEC  2023-02-12 00:00:00 NEC   ostr~   2.13   NA      NA      NA
10 2022-01-28_NDC  2022-01-28 00:00:00 NDC   ostr~   9.47    7.65    2.00    9.55
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code Your Figure

```r
ostracod_plot <- ggplot(data = ostracods,
                        mapping = aes(x = site_full,
                                      y = log_ave_lit)) +
  geom_jitter(height = 0,
              width = 0.2,
              shape = 21) +
```

```
  stat_summary(geom = "point",
               fun = median,
               color = "hotpink",
               size = 3) +
  labs(x = "Site",
       y = "Log +1 transformed average abundance/L",
       title = "NCOS Ostracod Abundance") +
  theme_bw() +
  scale_x_discrete(labels = label_wrap(width = 10))

ostracod_plot
```



d. Create a Multi Panel Figure

```
salinity_plot / (copepod_plot | ostracod_plot)
```